

# Computer Architecture

## Cache Memory

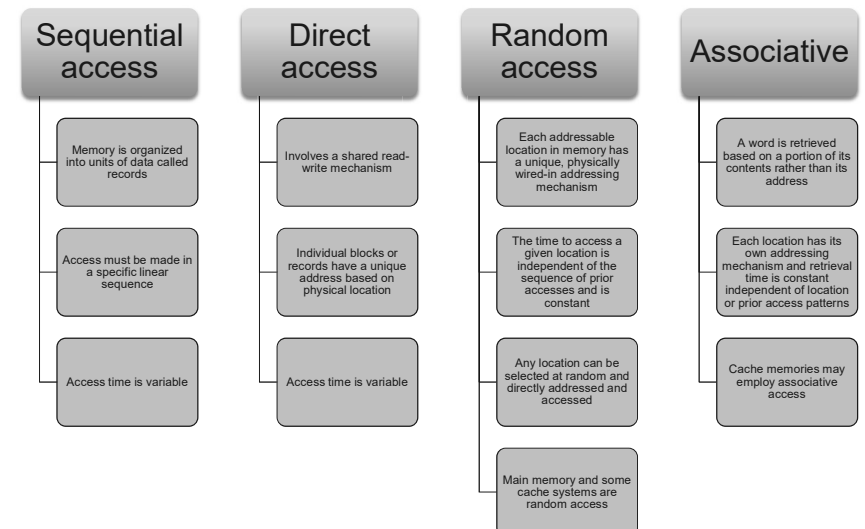
## Key Characteristics of Computer Memory Systems

|                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                 |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Location</b><br>Internal (e.g. processor registers, cache, main memory)<br>External (e.g. optical disks, magnetic disks, tapes)<br><b>Capacity</b><br>Number of words<br>Number of bytes<br><b>Unit of Transfer</b><br>Word<br>Block<br><b>Access Method</b><br>Sequential<br>Direct<br>Random<br>Associative | <b>Performance</b><br>Access time<br>Cycle time<br>Transfer rate<br><b>Physical Type</b><br>Semiconductor<br>Magnetic<br>Optical<br>Magneto-optical<br><b>Physical Characteristics</b><br>Volatile/nonvolatile<br>Erasable/nonerasable<br><b>Organization</b><br>Memory modules |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Characteristics of Memory Systems

- **Location**
  - Refers to whether memory is internal and external to the computer
  - Internal memory is often equated with main memory
  - Processor requires its own local memory, in the form of registers
  - Cache is another form of internal memory
  - External memory consists of peripheral storage devices that are accessible to the processor via I/O controllers
- **Capacity**
  - Memory is typically expressed in terms of bytes
- **Unit of transfer**
  - For internal memory the unit of transfer is equal to the number of electrical lines into and out of the memory module

## Method of Accessing Units of Data



## Capacity and Performance:

### The two most important characteristics of memory

### Three performance parameters are used:

#### Access time (latency)

- For random-access memory it is the time it takes to perform a read or write operation
- For non-random-access memory it is the time it takes to position the read-write mechanism at the desired location

#### Memory cycle time

- Access time plus any additional time required before second access can commence
- Additional time may be required for transients to die out on signal lines or to regenerate data if they are read destructively
- Concerned with the system bus, not the processor

#### Transfer rate

- The rate at which data can be transferred into or out of a memory unit
- For random-access memory it is equal to  $1/(\text{cycle time})$

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## Memory

- The most common forms are:
  - Semiconductor memory
  - Magnetic surface memory
  - Optical
  - Magneto-optical
- Several physical characteristics of data storage are important:
  - Volatile memory
    - Information decays naturally or is lost when electrical power is switched off
  - Nonvolatile memory
    - Once recorded, information remains without deterioration until deliberately changed
    - No electrical power is needed to retain information
  - Magnetic-surface memories
    - Are nonvolatile
  - Semiconductor memory
    - May be either volatile or nonvolatile
  - Nonerasable memory
    - Cannot be altered, except by destroying the storage unit
    - Semiconductor memory of this type is known as read-only memory (ROM)
- For random-access memory the organization is a key design issue
  - Organization refers to the physical arrangement of bits to form words

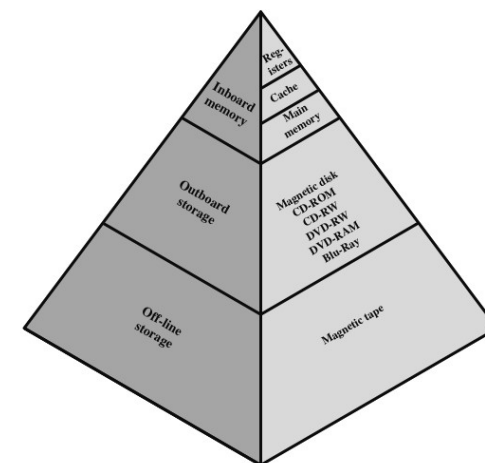
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## Memory Hierarchy

- Design constraints on a computer's memory can be summed up by three questions:
  - How much, how fast, how expensive
- There is a trade-off among capacity, access time, and cost
  - Faster access time, greater cost per bit
  - Greater capacity, smaller cost per bit
  - Greater capacity, slower access time
- The way out of the memory dilemma is not to rely on a single memory component or technology, but to employ a memory hierarchy

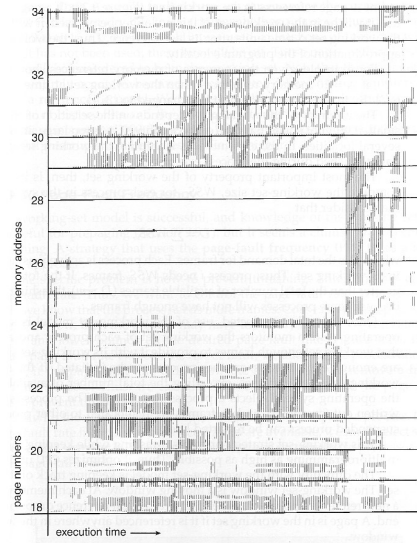
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## Memory Hierarchy - Diagram



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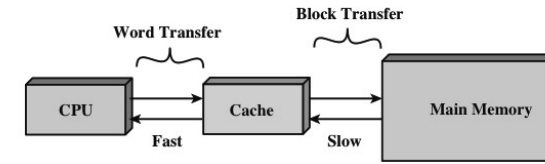
## Locality of reference



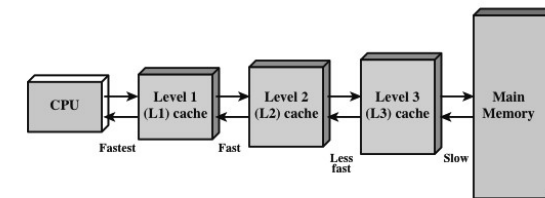
Abraham Silberschatz, Greg Gagne, and Peter Baer Galvin, "Operating System Concepts", 9th ed, Chapter 9

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## Cache and Main Memory



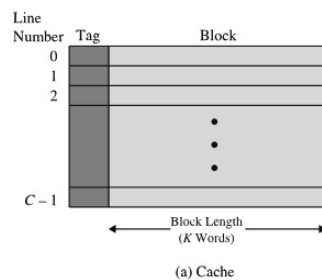
(a) Single cache



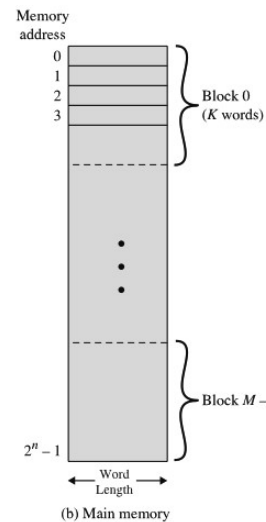
(b) Three-level cache organization

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## Cache/Main Memory Structure



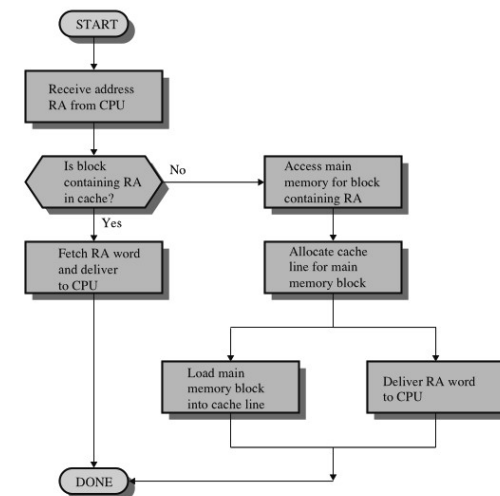
(a) Cache



(b) Main memory

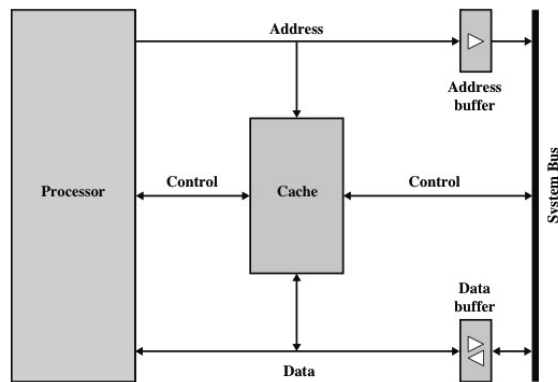
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## Cache Read Operation



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## Typical Cache Organization



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## Elements of Cache Design

|                              |                         |
|------------------------------|-------------------------|
| <b>Cache Addresses</b>       | <b>Write Policy</b>     |
| Logical                      | Write through           |
| Physical                     | Write back              |
| <b>Cache Size</b>            | <b>Line Size</b>        |
| <b>Mapping Function</b>      | <b>Number of caches</b> |
| Direct                       | Single or two level     |
| Associative                  | Unified or split        |
| Set Associative              |                         |
| <b>Replacement Algorithm</b> |                         |
| Least recently used (LRU)    |                         |
| First in first out (FIFO)    |                         |
| Least frequently used (LFU)  |                         |
| Random                       |                         |

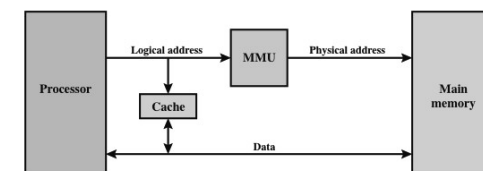
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## Cache Addresses

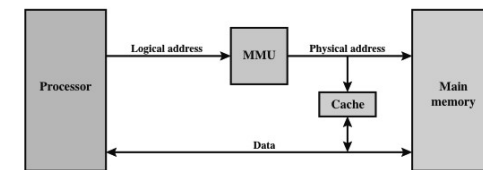
- Virtual memory
  - Facility that allows programs to address memory from a logical point of view, without regard to the amount of main memory physically available
  - When used, the address fields of machine instructions contain virtual addresses
  - For reads to and writes from main memory, a hardware memory management unit (MMU) translates each virtual address into a physical address in main memory

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## Logical and Physical Caches



(a) Logical Cache



(b) Physical Cache

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## Mapping Function

- Because there are fewer cache lines than main memory blocks, an algorithm is needed for mapping main memory blocks into cache lines
- Three techniques can be used:

### Direct

- The simplest technique
- Maps each block of main memory into only one possible cache line

### Associative

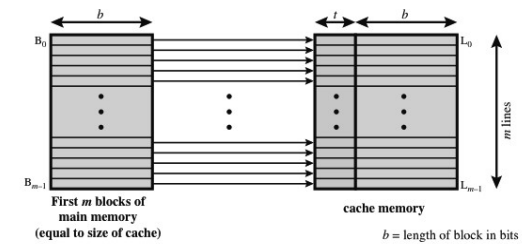
- Permits each main memory block to be loaded into any line of the cache
- The cache control logic interprets a memory address simply as a Tag and a Word field
- To determine whether a block is in the cache, the cache control logic must simultaneously examine every line's Tag for a match

### Set Associative

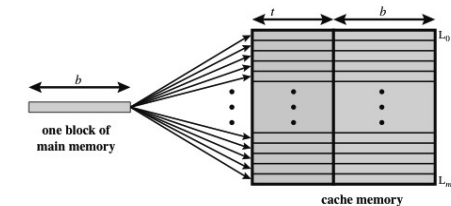
- A compromise that exhibits the strengths of both the direct and associative approaches while reducing their disadvantages

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## Direct Mapping



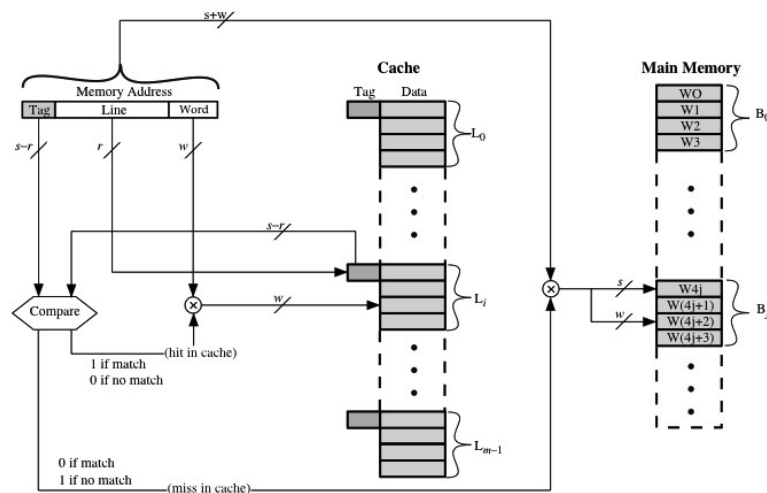
(a) Direct mapping



(b) Associative mapping

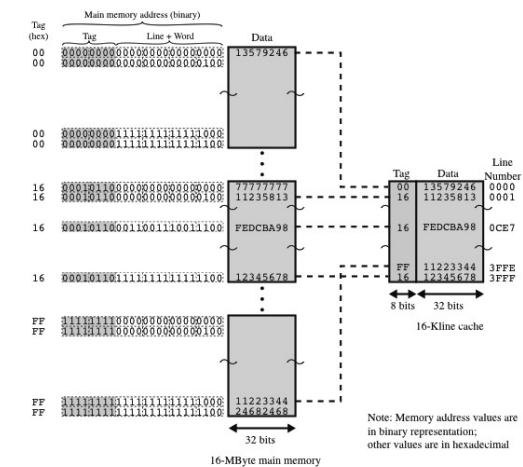
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## Direct Mapping Cache Organization



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## Direct Mapping Example



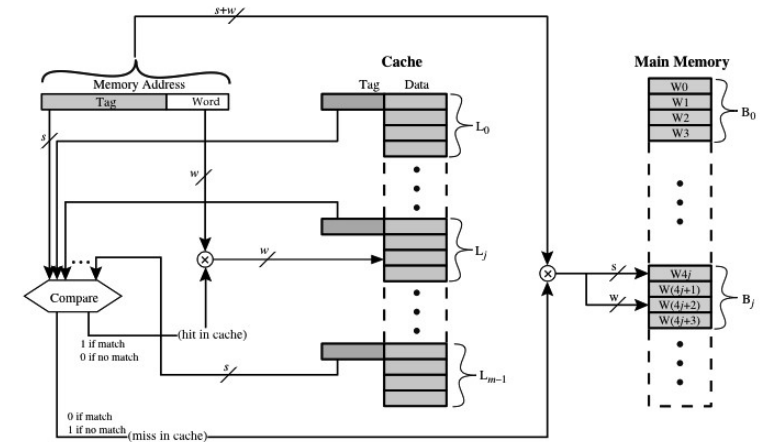
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## Direct Mapping Summary

- Address length =  $(s + w)$  bits
- Number of addressable units =  $2^{s+w}$  words or bytes
- Block size = line size =  $2^w$  words or bytes
- Number of blocks in main memory =  $2^{s+w}/2^w = 2^s$
- Number of lines in cache =  $m = 2^r$
- Size of tag =  $(s - r)$  bits

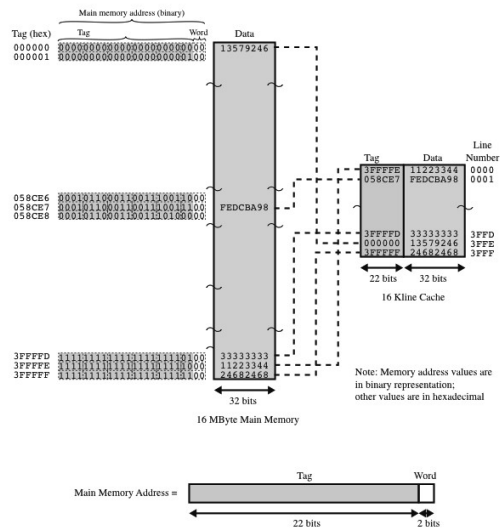
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## Fully Associative Cache Organization



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## Associative Mapping Example



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## Associative Mapping Summary

- Address length =  $(s + w)$  bits
- Number of addressable units =  $2^{s+w}$  words or bytes
- Block size = line size =  $2^w$  words or bytes
- Number of blocks in main memory =  $2^{s+w}/2^w = 2^s$
- Number of lines in cache = undetermined
- Size of tag =  $s$  bits

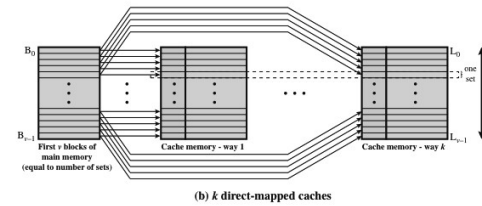
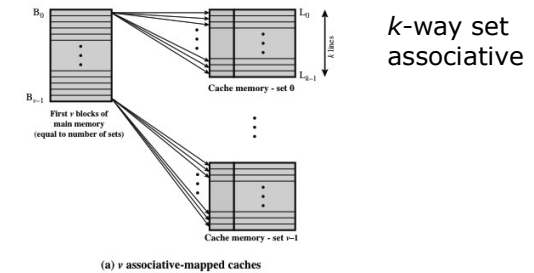
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## Set Associative Mapping

- Compromise that exhibits the strengths of both the direct and associative approaches while reducing their disadvantages
- Cache consists of a number of sets
- Each set contains a number of lines
- A given block maps to any line in a given set
- e.g. 2 lines per set
  - 2 way associative mapping
  - A given block can be in one of 2 lines in only one set

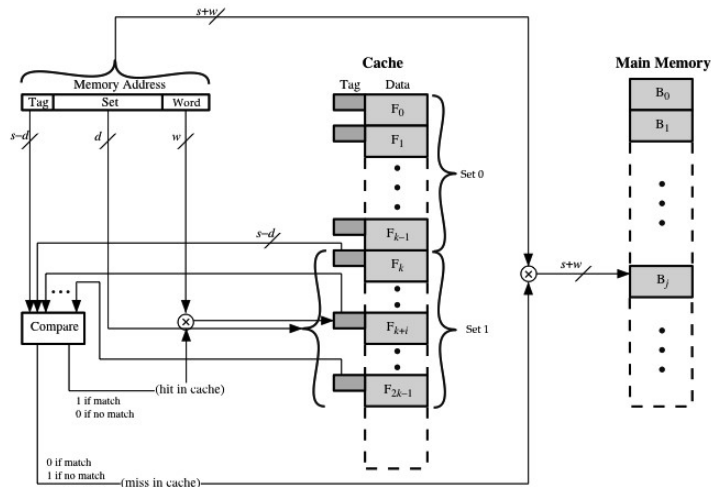
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## Mapping From Main Memory to Cache:



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## k-Way Set Associative Cache Organization



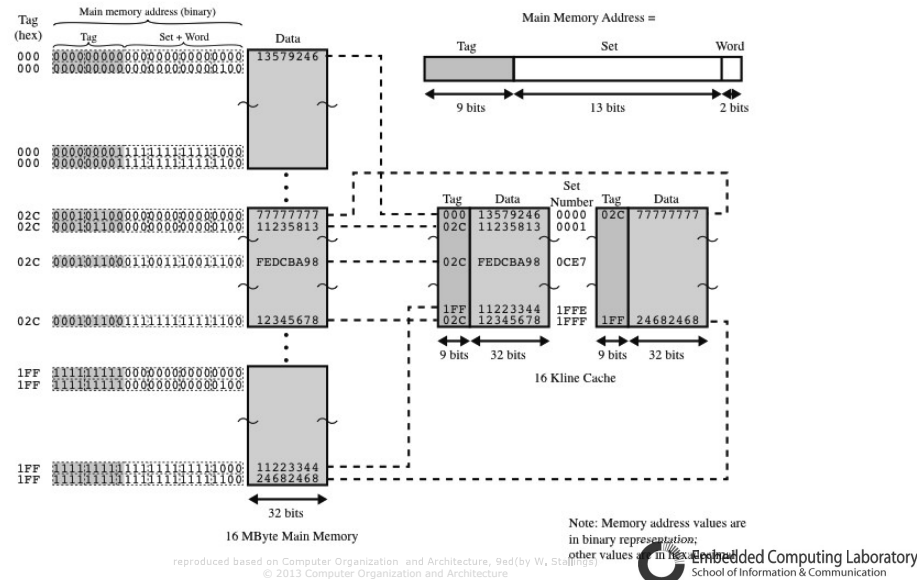
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## Set Associative Mapping Summary

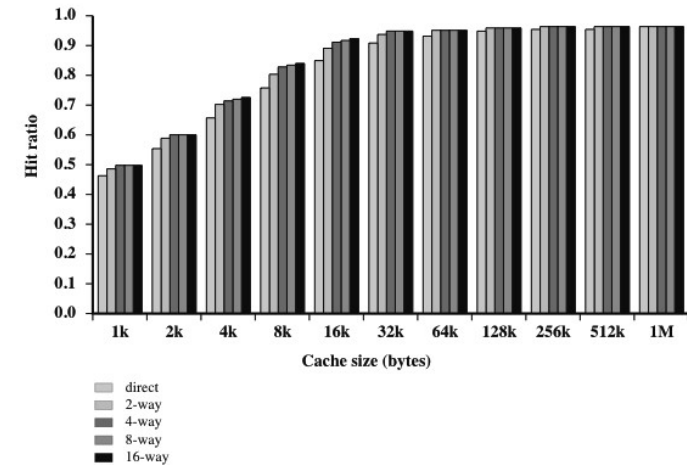
- Address length =  $(s + w)$  bits
- Number of addressable units =  $2^{s+w}$  words or bytes
- Block size = line size =  $2^w$  words or bytes
- Number of blocks in main memory =  $2^{s+w}/2^w = 2^s$
- Number of lines in set =  $k$
- Number of sets =  $v = 2^d$
- Number of lines in cache =  $m = kv = k * 2^d$
- Size of cache =  $k * 2^{d+w}$  words or bytes
- Size of tag =  $(s - d)$  bits

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## 2-way Set Associative Mapping Example



## Varying Associativity Over Cache Size



## Replacement Algorithms

- Once the cache has been filled, when a new block is brought into the cache, one of the existing blocks must be replaced
- For direct mapping there is only one possible line for any particular block and no choice is possible
- For the associative and set-associative techniques a replacement algorithm is needed
- To achieve high speed, an algorithm must be implemented in hardware

## The four most common replacement algorithms are:

- Least recently used (LRU)
  - Most effective
  - Replace that block in the set that has been in the cache longest with no reference to it
  - Because of its simplicity of implementation, LRU is the most popular replacement algorithm
- First-in-first-out (FIFO)
  - Replace that block in the set that has been in the cache longest
  - Easily implemented as a round-robin or circular buffer technique
- Least frequently used (LFU)
  - Replace that block in the set that has experienced the fewest references
  - Could be implemented by associating a counter with each line



## Write Policy

When a block that is resident in the cache is to be replaced there are two cases to consider:

If the old block in the cache has not been altered then it may be overwritten with a new block without first writing out the old block

If at least one write operation has been performed on a word in that line of the cache then main memory must be updated by writing the line of cache out to the block of memory before bringing in the new block

There are two problems to contend with:

More than one device may have access to main memory

A more complex problem occurs when multiple processors are attached to the same bus and each processor has its own local cache - if a word is altered in one cache it could conceivably invalidate a word in other caches

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## Write Through vs. Write Back

- Write through
  - Simplest technique
  - All write operations are made to main memory as well as to the cache
  - The main disadvantage of this technique is that it generates substantial memory traffic and may create a bottleneck
- Write back
  - Minimizes memory writes
  - Updates are made only in the cache
  - Portions of main memory are invalid and hence accesses by I/O modules can be allowed only through the cache
  - This makes for complex circuitry and a potential bottleneck

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## Line Size

When a block of data is retrieved and placed in the cache not only the desired word but also some number of adjacent words are retrieved

As the block size increases more useful data are brought into the cache

Two specific effects come into play:  
 • Larger blocks reduce the number of blocks that fit into a cache  
 • As a block becomes larger each additional word is farther from the requested word

As the block size increases the hit ratio will at first increase because of the principle of locality

The hit ratio will begin to decrease as the block becomes bigger and the probability of using the newly fetched information becomes less than the probability of reusing the information that has to be replaced

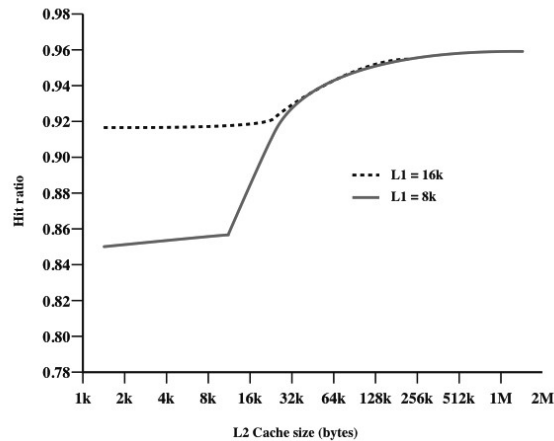
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## Multilevel Caches

- As logic density has increased it has become possible to have a cache on the same chip as the processor
- The on-chip cache reduces the processor's external bus activity and speeds up execution time and increases overall system performance
  - When the requested instruction or data is found in the on-chip cache, the bus access is eliminated
  - On-chip cache accesses will complete appreciably faster than would even zero-wait state bus cycles
  - During this period the bus is free to support other transfers
- Two-level cache:
  - Internal cache designated as level 1 (L1)
  - External cache designated as level 2 (L2)
- Potential savings due to the use of an L2 cache depends on the hit rates in both the L1 and L2 caches
- The use of multilevel caches complicates all of the design issues related to caches, including size, replacement algorithm, and write policy

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## Hit Ratio (L1 & L2) for 8 Kbyte and 16 Kbyte L1



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## Unified Versus Split Caches

- Has become common to split cache:
  - One dedicated to instructions
  - One dedicated to data
  - Both exist at the same level, typically as two L1 caches
- Advantages of unified cache:
  - Higher hit rate
    - Balances load of instruction and data fetches automatically
    - Only one cache needs to be designed and implemented
- Trend is toward split caches at the L1 and unified caches for higher levels
- Advantages of split cache:
  - Eliminates cache contention between instruction fetch/decode unit and execution unit
    - Important in pipelining

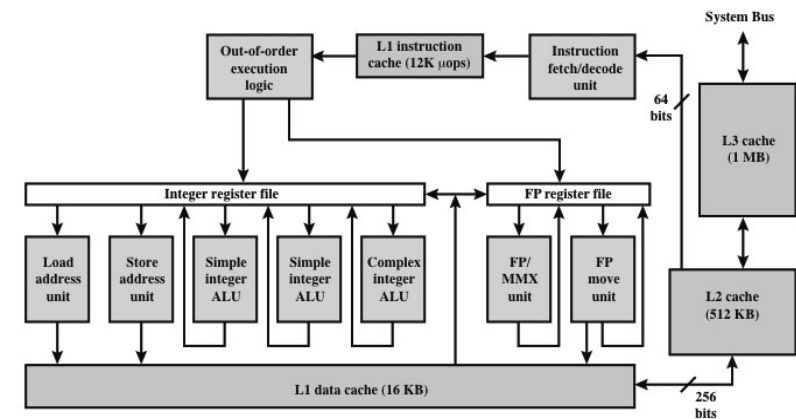
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## Intel CPU Cache Evolution

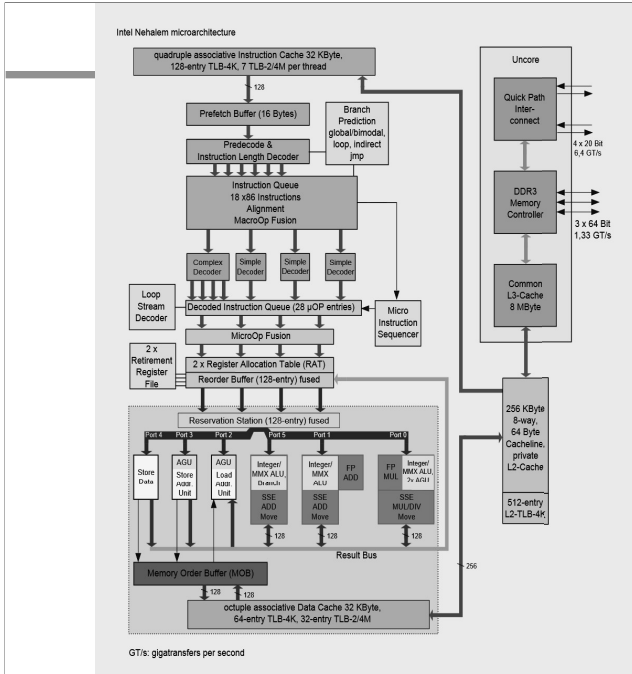
| Problem                                                                                                                                                                                                               | Solution                                                                                                                               | Processor on which Feature First Appears |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| External memory slower than the system bus.                                                                                                                                                                           | Add external cache using faster memory technology.                                                                                     | 386                                      |
| Increased processor speed results in external bus becoming a bottleneck for cache access.                                                                                                                             | Move external cache on-chip, operating at the same speed as the processor.                                                             | 486                                      |
| Internal cache is rather small, due to limited space on chip                                                                                                                                                          | Add external L2 cache using faster technology than main memory                                                                         | 486                                      |
| Contention occurs when both the Instruction Prefetcher and the Execution Unit simultaneously require access to the cache. In that case, the Prefetcher is stalled while the Execution Unit's data access takes place. | Create separate data and instruction caches.                                                                                           | Pentium                                  |
| Increased processor speed results in external bus becoming a bottleneck for L2 cache access.                                                                                                                          | Create separate back-side bus that runs at higher speed than the main (front-side) external bus. The BSB is dedicated to the L2 cache. | Pentium Pro                              |
|                                                                                                                                                                                                                       | Move L2 cache on to the processor chip.                                                                                                | Pentium II                               |
| Some applications deal with massive databases and must have rapid access to large amounts of data. The on-chip caches are too small.                                                                                  | Add external L3 cache.                                                                                                                 | Pentium III                              |
|                                                                                                                                                                                                                       | Move L3 cache on-chip.                                                                                                                 | Pentium 4                                |

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## Pentium 4 Block Diagram



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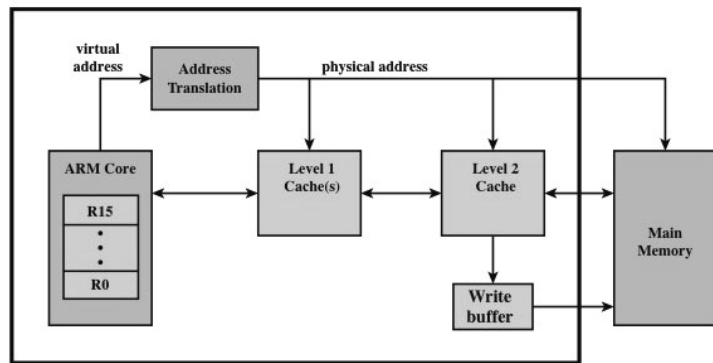


## Intel Nehalem Architecture (i-7 core in 2008)

## ARM Cache Features

| Core            | Cache Type | Cache Size (kB) | Cache Line Size (words) | Associativity | Location | Write Buffer Size (words) |
|-----------------|------------|-----------------|-------------------------|---------------|----------|---------------------------|
| ARM720T         | Unified    | 8               | 4                       | 4-way         | Logical  | 8                         |
| ARM920T         | Split      | 16/16 D/I       | 8                       | 64-way        | Logical  | 16                        |
| ARM926EJ-S      | Split      | 4-128/4-128 D/I | 8                       | 4-way         | Logical  | 16                        |
| ARM1022E        | Split      | 16/16 D/I       | 8                       | 64-way        | Logical  | 16                        |
| ARM1026EJ-S     | Split      | 4-128/4-128 D/I | 8                       | 4-way         | Logical  | 8                         |
| Intel StrongARM | Split      | 16/16 D/I       | 4                       | 32-way        | Logical  | 32                        |
| Intel Xscale    | Split      | 32/32 D/I       | 8                       | 32-way        | Logical  | 32                        |
| ARM1136-JF-S    | Split      | 4-64/4-64 D/I   | 8                       | 4-way         | Physical | 32                        |

## ARM Cache and Write Buffer Organization



## Summary

- Characteristics of Memory Systems
  - Location
  - Capacity
  - Unit of transfer
- Memory Hierarchy
  - How much?
  - How fast?
  - How expensive?
- Elements of cache design
  - Cache addresses
  - Cache size
  - Mapping function
  - Replacement algorithms
  - Write policy
  - Line size
  - Number of caches

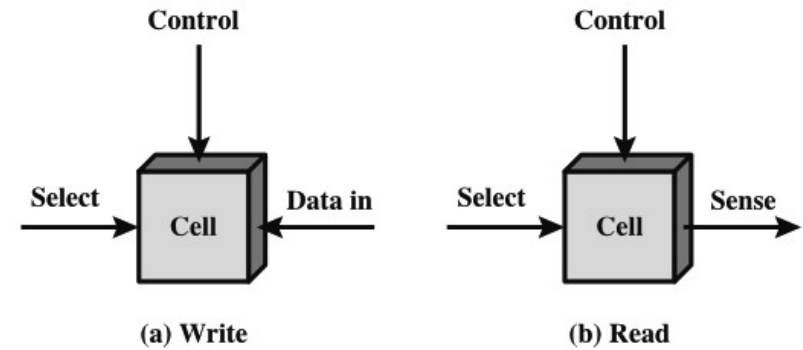


Image from <http://www.promotionalpro.com/blog/>

# Computer Architecture

## Internal Memory

## Memory Cell Operation

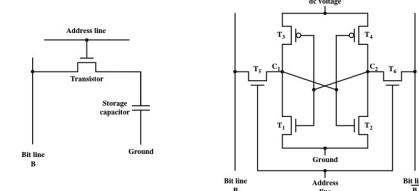


## Semiconductor Memory Types

| Memory Type                         | Category           | Erasure                   | Write Mechanism | Volatility  |
|-------------------------------------|--------------------|---------------------------|-----------------|-------------|
| Random-access memory (RAM)          | Read-write memory  | Electrically, byte-level  | Electrically    | Volatile    |
| Read-only memory (ROM)              | Read-only memory   | Not possible              | Masks           | Nonvolatile |
| Programmable ROM (PROM)             |                    |                           |                 |             |
| Erasable PROM (EPROM)               | Read-mostly memory | UV light, chip-level      | Electrically    |             |
| Electrically Erasable PROM (EEPROM) |                    | Electrically, byte-level  |                 |             |
| Flash memory                        |                    | Electrically, block-level |                 |             |

## RAM (Random Access Memory)

- Dynamic RAM (DRAM)
  - Made with cells that store data as charge on capacitors
  - Presence or absence of charge in a capacitor is interpreted as a binary 1/0
  - Requires periodic charge refreshing to maintain data storage
  - The term dynamic refers to tendency of the stored charge to leak away, even with power continuously applied
- Static RAM (SRAM)
  - Digital device that uses the same logic elements used in the processor
  - Binary values are stored using traditional flip-flop logic gate configurations
  - Will hold its data as long as power is supplied to it



## Read Only Memory (ROM)

- Contains a permanent pattern of data that cannot be changed or added to
- No power source is required to maintain the bit values in memory
- Data or program is permanently in main memory and never needs to be loaded from a secondary storage device
- Data is actually wired into the chip as part of the fabrication process
  - Disadvantages of this:
    - No room for error, if one bit is wrong the whole batch of ROMs must be thrown out
    - Data insertion step includes a relatively large fixed cost

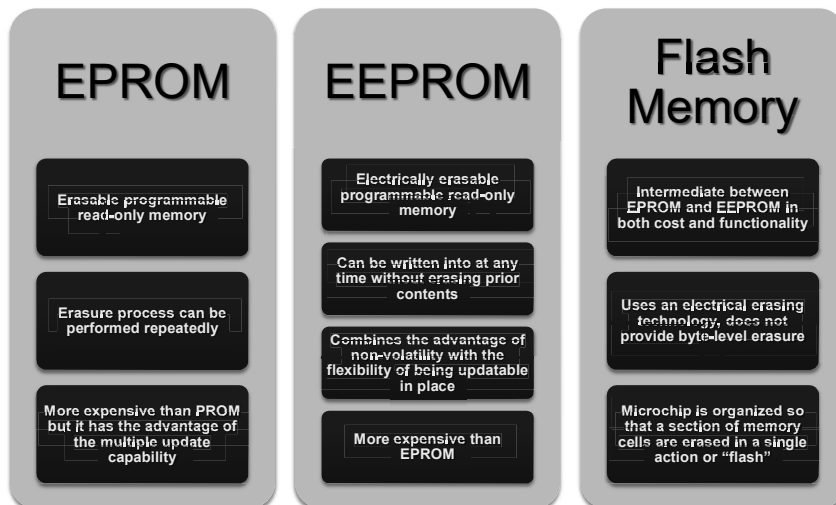
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## Programmable ROM (PROM)

- Less expensive alternative
- Nonvolatile and may be written into only once
- Writing process is performed electrically and may be performed by supplier or customer at a time later than the original chip fabrication
- Special equipment is required for the writing process
- Provides flexibility and convenience
- Attractive for high volume production runs

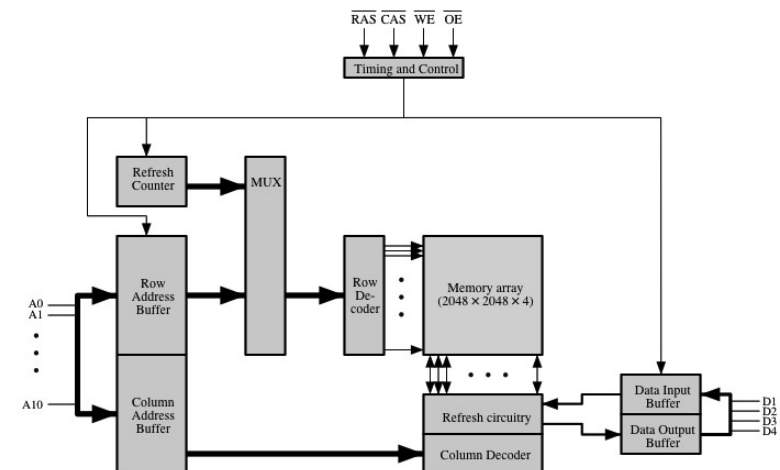
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## Read-Mostly Memory



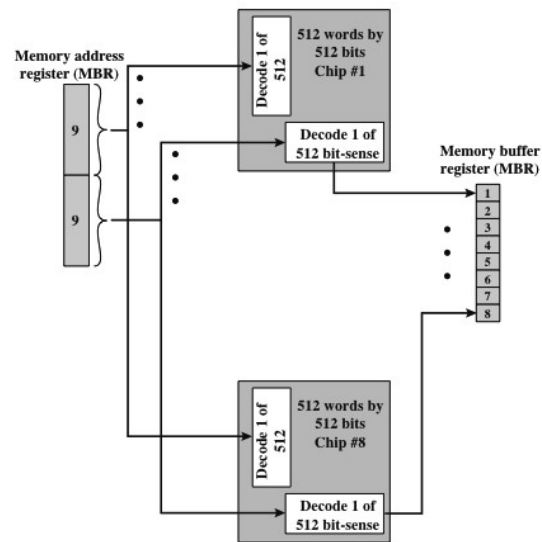
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## Typical 16 Mb DRAM (4M x 4)



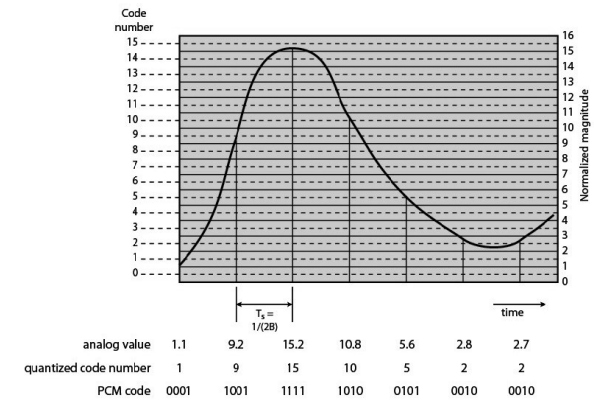
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## Memory Organization Example (256 KByte)



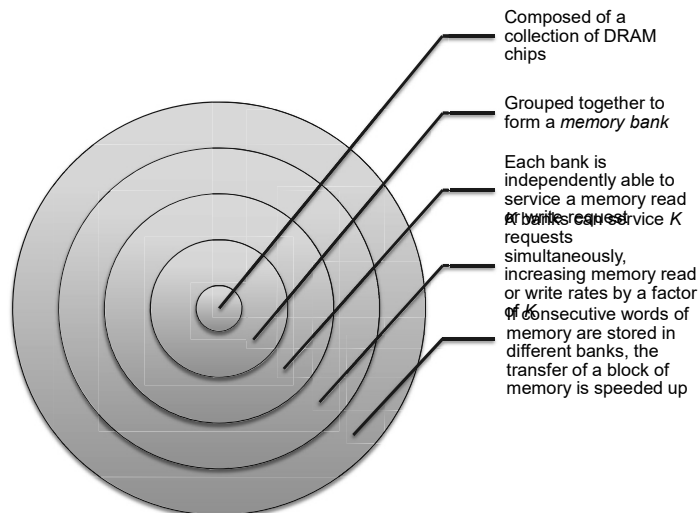
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## 1MByte Module Organization



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## Interleaved Memory



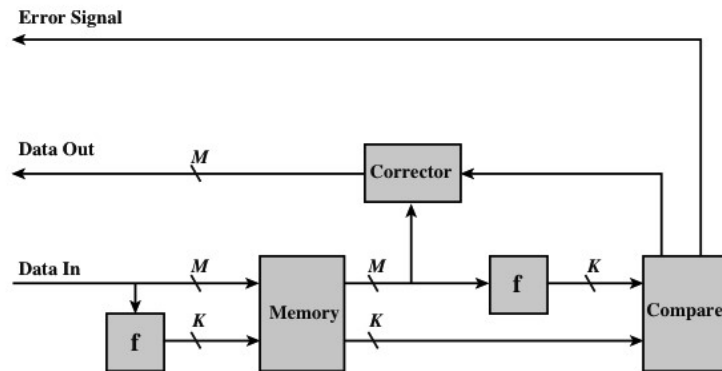
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## Error Correction

- Hard Failure
  - Permanent physical defect
  - Memory cell or cells affected cannot reliably store data but become stuck at 0 or 1 or switch erratically between 0 and 1
  - Can be caused by:
    - Harsh environmental abuse
    - Manufacturing defects
    - Wear
- Soft Error
  - Random, non-destructive event that alters the contents of one or more memory cells
  - No permanent damage to memory
  - Can be caused by:
    - Power supply problems
    - Alpha particles

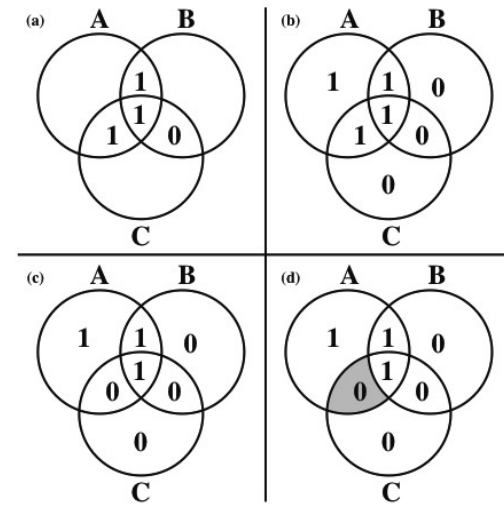
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## Error Correcting Code Function



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## Hamming Error Correcting Code



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## Layout of Data Bits and Check Bits

| Bit Position    | 12   | 11   | 10   | 9    | 8    | 7    | 6    | 5    | 4    | 3    | 2    | 1    |
|-----------------|------|------|------|------|------|------|------|------|------|------|------|------|
| Position Number | 1100 | 1011 | 1010 | 1001 | 1000 | 0111 | 0110 | 0101 | 0100 | 0011 | 0010 | 0001 |
| Data Bit        | D8   | D7   | D6   | D5   |      | D4   | D3   | D2   |      | D1   |      |      |
| Check Bit       |      |      |      |      | C8   |      |      |      | C4   |      | C2   | C1   |

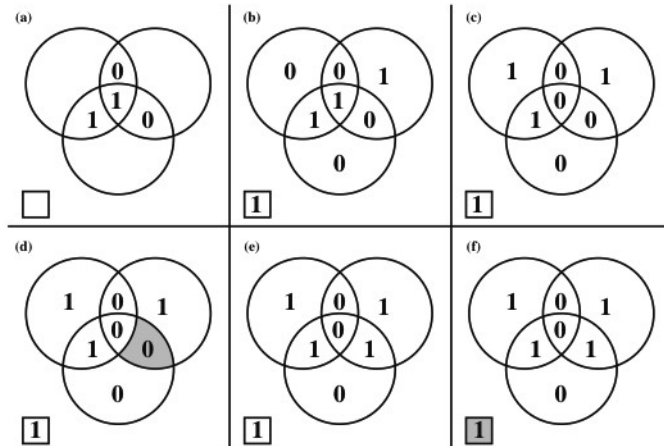
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## Check Bit Calculation

| Bit position    | 12   | 11   | 10   | 9    | 8    | 7    | 6    | 5    | 4    | 3    | 2    | 1    |
|-----------------|------|------|------|------|------|------|------|------|------|------|------|------|
| Position number | 1100 | 1011 | 1010 | 1001 | 1000 | 0111 | 0110 | 0101 | 0100 | 0011 | 0010 | 0001 |
| Data bit        | D8   | D7   | D6   | D5   |      | D4   | D3   | D2   |      | D1   |      |      |
| Check bit       |      |      |      |      | C8   |      |      |      | C4   |      | C2   | C1   |
| Word stored as  | 0    | 0    | 1    | 1    | 0    | 1    | 0    | 0    | 1    | 1    | 1    | 1    |
| Word fetched as | 0    | 0    | 1    | 1    | 0    | 1    | 1    | 0    | 1    | 1    | 1    | 1    |
| Position Number | 1100 | 1011 | 1010 | 1001 | 1000 | 0111 | 0110 | 0101 | 0100 | 0011 | 0010 | 0001 |
| Check Bit       |      |      |      |      | 0    |      |      |      | 0    |      | 0    | 1    |

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## Hamming SEC-DED Code



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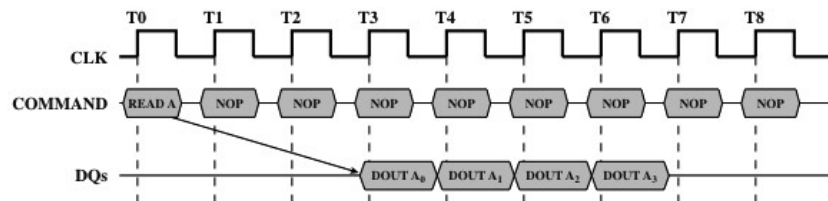
## Advanced DRAM Organization

- One of the most critical system bottlenecks when using high-performance processors is the interface to main internal memory
- The traditional DRAM chip is constrained both by its internal architecture and by its interface to the processor's memory bus
- A number of enhancements to the basic DRAM architecture have been explored:

|       | Clock Frequency (MHz) | Transfer Rate (GB/s) | Access Time (ns) | Pin Count |
|-------|-----------------------|----------------------|------------------|-----------|
| SDRAM | 166                   | 1.3                  | 18               | 168       |
| DDR   | 200                   | 3.2                  | 12.5             | 184       |
| RDRAM | 600                   | 4.8                  | 12               | 162       |

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## SDRAM Read Timing



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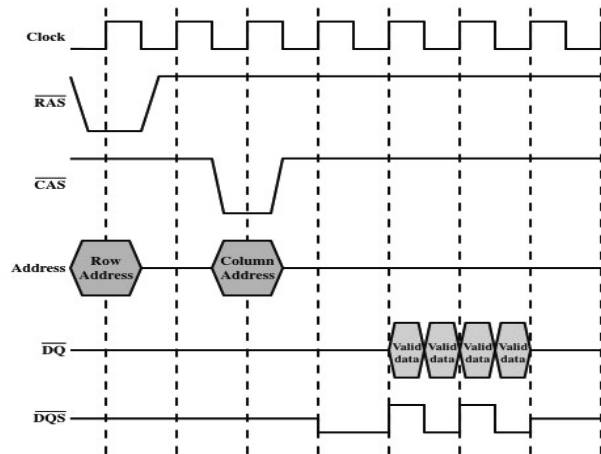
## Double Data Rate SDRAM (DDR SDRAM)

- SDRAM can only send data once per bus clock cycle
- Double-data-rate SDRAM can send data twice per clock cycle, once on the rising edge of the clock pulse and once on the falling edge
- Developed by the JEDEC Solid State Technology Association (Electronic Industries Alliance's semiconductor-engineering-standardization body)

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## DDR SDRAM Read Timing



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## Cache DRAM (CDRAM)

- Developed by Mitsubishi
- Integrates a small SRAM cache onto a generic DRAM chip
- SRAM on the CDRAM can be used in two ways:
  - It can be used as a true cache consisting of a number of 64-bit lines
    - Cache mode of the CDRAM is effective for ordinary random access to memory
  - Can also be used as a buffer to support the serial access of a block of data

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## Summary

- Semiconductor main memory
  - Organization
  - DRAM and SRAM
  - Types of ROM
  - Chip logic
  - Chip packaging
  - Module organization
  - Interleaved memory
- Error correction
  - Hard failure
  - Soft error
  - Hamming code
- Advanced DRAM organization
  - Synchronous DRAM
  - DDR SDRAM
  - Cache DRAM

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